BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Lahore Board 10th - 2022) (Paper - I) (Objective Type) (Group - I)

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer -book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. What does the term e-mail stand for:

(A) Emergency mail
(B) Electronic mail
(C) Extra mail
(D) External mail

Q2. Michael Faraday belonged to:

(A) British
(B) USA
(C) KSA
(D) Russia

Q3. Release of energy by the sun is due to:

(A) Nuclear fission
(B) Nuclear fusion
(C) Burning of gases
(D) Chemical reaction

Q4. Which of the following characteristics of a wave is independent of the others:

(A) Speed
(B) Frequency
(C) Amplitude
(D) Wavelength

Q5. Charge on alpha particles is:

(A) 2e
(B) 3e
(C) 4e
(D) 5e

Q6. For a normal person, audible frequency range for sound wave lies between:

(A) 10Hz - 10KHz
(B) 20Hz - 20KHz
(C) 25Hz - 25KHz
(D) 30Hz - 30KHz

Q7. How many components a CBIS has:

(A) 2
(B) 3
(C) 5
(D) 6

Q8. The instrument used to detect the nature of charge is:

(A) Stroboscope
(B) Electroscope
(C) Spectroscope
(D) Microscope

Q9. An object is 14cm in front of a convex mirror. The image is 5.8cm behind the mirror. What is the focal length of the mirror:

(A) -4.1cm
(B) -8.2cm
(C) -9.9cm
(D) -20cm

Q10. The output of the NAND gate is written as:

(A) X= A+B
(B) X= A-B
(C) X= A.B
(D) X= overline(A+B)

Q11. One watt is equal to:

(A) JS
(B) JS^{-1}
(C) J
(D) Sec

Q12. AND gates can be form by using two:

(A) NAND gates
(B) OR gates
(C) NOR gates
(D) NOT gates

Physics (Lahore Board 10th - 2022) (Paper - II) (Subjective Type) (Group - I)

Time Allowed: 1.45 Min. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What is the cause of flow of current in electrolytes?

(ii) State Ohm's law and write its equation.

(iii) Can magnetic poles be seperated? How?

(iv) What is right hand rule to determine the magnetic poles of a current carrying coil?

(v) Write working principle of electric motor?

(vi) Is it possible for an element to have different type of atoms? Explain.

(vii) Give two examples of the production of radio isotopes?

(viii) Find the number of protons and neutrons in the nuclide defined by $^{13}_{6}X$

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Define loudness. On what factors does the loudness of sound depend?

(ii) Define sound waves and write an example?

(iii) Enlist four uses of capacitors?

(iv) Define columb's law and write its equation?

(v) Define Internet and write two services of Internet?

(vi) What is difference between RAM and ROM memories?

(vii) Calculate the frequency of a sound wave of speed $340ms^{-1}$ and wavelength 0.5m.

(viii) What are Browsers? Give their two examples.

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What is meant by diffraction of waves?

(ii) Define spring constant and write its formula.

(iii) Describe the conditions for total internal reflection.

(iv) Differentiate between core and cladding of an optical fibre.

(v) What is meant by near point of eye?

(vi) How the filament is heated in an oscilloscope and why it is heated? (reason)

(vii) Draw the symbolic diagram of OR gate and write down its truth table.

(viii) If f = 2Hz and λ = 0.2m then calculate the speed and time period?

Physics (Lahore Board 10th - 2022) (Paper - II) (PART - II) (Group - I)

Time Allowed: 1.45 Min. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). What is electromagnetic induction? Explain it with activity.

Q5(b). Ashes from a campfire deep in a cave show carbon-14 activity of only one-eighth the activity of fresh wood. How long ago was the campfire made?

Q6(a). Explain the electrostatic by example of electrostatic powder spray painting.

Q6(b). A doctor counts 72 heartbeats in one minute. Calculate the frequency and period of the heartbeats.

Q7(a). Indicate the objects of cathode ray oscilloscope. Show its different parts with diagram.

Q7(b). A pendulum of length 0.99m is taken to the moon by an astronaut. The period of the pendulum is 4.9s. What is the value of 'g' on the surface of moon?

GROUP – II

Physics (Lahore Board 10th - 2022) (Paper - I) (Objective Type) (Group - II)

Time Allowed: 15 Min. | **Maximum Marks:** 12

> Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer -book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. If the length of the pendulum is doubled, its time period will be:

(A) √2T
(B) 2T
(C) T/2
(D) T/√2

Q2. The characteristic of sound by which we can distinguish between two sounds of same loudness and pitch is called:

(A) Pitch
(B) Quality
(C) Loudness
(D) Intensity

Q3. The image formed by concave mirror is:

(A) Real
(B) Virtual
(C) Real or Virtual
(D) Real and Virtual

Q4. Two capacitors of capacitance 12μf and 6μf are connected in parallel with 12V battery, The equivalent capacitance is:

(A) 6μf
(B) 12μf
(C) 1.2μf
(D) 18μf

Q5. What is the voltage across a 3Ω resistor when 6A of current passes through it:

(A) 36V
(B) 9V
(C) 18V
(D) 2V

Q6. The application of mutual induction is:

(A) D.C. motor
(B) Transformer
(C) Relay
(D) A.C. motor

Q7. Boolean expression for NAND gate is:

(A) X = overline(A.B)
(B) X = A.B
(C) X = overline(A+B)
(D) X = A+B

Q8. Number of components of computer based information system is:

(A) 3
(B) 4
(C) 5
(D) 6

Q9. Which of the following radiations has more penetrating power:

(A) Gamma rays
(B) Alpha particles
(C) Beta particles
(D) All have same

Q10. The number of protons in $^{238}_{92}U$ are:

(A) 146
(B) 238
(C) 196
(D) 92

Q11. The current of 3A is flowing through a wire for one minute. What is charge in wire:

(A) 3C
(B) 180C
(C) $180\times10^{-3}C$
(D) 20C

Q12. The unit of sound intensity is:

(A) Wm^{-2}
(B) Wm^{-1}
(C) Wm^{2}
(D) Wm

Physics (Lahore Board 10th - 2022) (Paper - I) (Subjective Type) (Group - II)

Time Allowed: 1.45 Min. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Define electric power and write the name of its unit.

(ii) Define specific resistance and write its formula also.

(iii) Write equation of equivalent resistance of a parallel combination of resistors and draw its circuit diagram.

(iv) Define transformer and write its principle.

(v) Write two factors which affects induced e.m.f.

(vi) How the magnetic force acting on a current carrying wire can be increased? Write two methods.

(vii) What is meant by background radiations, write and example also.

(viii) Write two characteristics of beta particles.

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What is the necessary condition for the production of sound?

(ii) On what factor does the loudness depend?

(iii) What is unit of loudness and define it.

(iv) State Coulomb's law of electrostatics and write its mathemetical form.

(v) What is meant by the term electric field and electric intensity?

(vi) Is electric intensity a vector quantity? What will be its direction.

(vii) Contrast between hardware and software.

(viii) Which two types of waves are used in modern telecommunication?

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) With the help of diagram show the kinetic and potential energy values at different positions in mass spring system.

(ii) How damping progressively reduces the amplitude of oscillation?

(iii) What is the nature of image formed by the plane mirror?

(iv) Why the focal length of convex mirror is taken negative?

(v) What is the difference between principal axis and principal focus?

(vi) Draw a graph of analogue and digital signals.

(vii) Draw circuit diagram of OR operation and AND operation.

(viii) Make the truth table of OR gate.

Physics (Lahore Board 10th - 2022) (Paper - I) (PART - II) (Group - II)

Time Allowed: 1.45 Min. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5. (a) What is e.m.f.? Explain the method to measure e.m.f. with diagram. (b) A step-down transformer has turns ration 100:1. An ac voltage of amplitude 170V is

applied to the primary. If the current in the primary is 1.0 m,A what is the current in the secondary?

Q6. (a) How optical fibres are used as channel of transmission of light signals? Explain it with diagram. (b) The electric potential at a point in an electric field is 10^4 V. If a charge of 100μC is brought from infinity to this point what would be the amount of work done on it?

Q7. (a) Define simple harmonic motion and prove with example that mass attached to spring shows simple harmonic motion. Also draw its label diagram. (b) An object 4 cm high is placed at a distance of 12cm from a convex lens of focal length 8 cm. Calculate the position and size of the image.